Ordovician paleomagnetic results from the St. George and Table Head carbonates of western Newfoundland

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We have studied 85 limestone and dolostone samples from 21 sites of the St. George and Table Head groups (Lower and Middle Ordovician) on Port au Port Peninsula (Humber Zone). Their gentle ($\sim 15°$) tilt is probably Devonian. A steep viscous component dominant in the natural remanence is removed by thermal and (less efficiently) alternating-field demagnetization. In about 60% of all samples, the thermal treatment yielded a southeasterly reverse component (''A''), which typically remained blocked to 450–500°C and then decayed to noise. Mean A component directions calculated from Zijderveld plots and referred to bedding are D, $I = 148.5°$, $+33.6°$ for $N = 9$ St. George sites and D, $I = 150.0°$, $I = +40.7°$ for $N = 5$ Table Head sites. Corresponding antipoles are 17.5°N, 152.3°E, dp, dm = 2.8°, 4.9° and 13.4°N, 149.2°E, dp, dm = 2.9°, 4.7°, respectively. We interpret these results as representative of the geomagnetic field in the Early and mid-Ordovician. In 14 samples, mainly from the older St. George strata, thermal treatment yielded a reverse (''B'') component with mean direction D, $I = 167.9°$, $-9.2°$, corresponding to an antipole at 44.9°N, 138.2°E, dp, dm = 3.6°, 7.1°. We interpret this as a late Paleozoic overprint in diagenetic hematite. The 30° anticlockwise rotation of Newfoundland proposed by Wegener is not supported by our data.

Nous avons étudié 85 échantillons de calcaire et de dolomie prélevés à 21 sites dans les groupes de St. George et de Table Head (Ordovicien inférieur et moyen) sur la péninsule de Port au Port (zone Humber). Leur faible inclinaison ($\sim 15°$) est probablement due à un événement dévonien. Une composante visqueuse, abrupte et dominante, dans la rémanence naturelle est éliminée par les traitements de désaimantation thermique et par champ alternatif. Le traitement de désaimantation thermique a fourni une composante sud-est inversée («A») pour approximativement 60% de tous les échantillons, laquelle est demeurée temporairement et typiquement bloquée à 450–500°C pour ensuite s'effacer graduellement et ne créer qu'une perturbation superficielle. Les directions moyennes de la composante A calculées avec les courbes de Zijderveld et relatives à la stratification sont D, $I = 148{,}5°$, $+33{,}6°$, $N = 9$ pour les sites de St. George et D, $I = 150{,}0°$, $+40{,}7°$, $N = 5$ pour les sites de Table Head. Les antipôles correspondants sont 17,5°N, 152,3°E, dp, dm = 2,8°, 4,9° et 13,4°N, 149,2°E, dp, dm = 2,9°, 4,7°, respectivement. Nous considérons que ces résultats sont représentatifs du champ géomagnétique à l'Ordovicien inférieur et moyen. Le traitement thermique de 14 échantillons, particulièrement ceux des strates plus âgées de St. George, ont fourni une composante inverse («B») avec la direction moyenne D, $I = 167{,}9°$, $-9{,}2°$, correspondant à un antipôle à 44,9°N, 138,2°E, dp, dm = 3,6°, 7,1°. Nous croyons que cette composante B est due à une surimpression paléozoïque dans l'hématite diagénétique. Nos résultats ne sont pas en accord avec la rotation de Terre-Neuve de 30° dans le sens inverse des aiguilles d'une montre proposée par Wegener.

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Introduction

The objective of this work was to obtain reliable directions of the Ordovician geomagnetic field relative to western Newfoundland. This is the first reported part of a wider paleomagnetic study of Cambrian–Ordovican carbonate rocks on the Western Platform of Newfoundland. Such investigations are important for a number of reasons. The region formed the eastern margin of cratonic North America (Williams and Stevens 1974), and it has a nearly complete stratigraphic record of the early Paleozoic. This provides an excellent setting for paleomagnetic comparisons with other parts of the Appalachian–Caledonian belt, for example as a test of the proto-Atlantic hypothesis (Wilson 1966) or for comparing cratonic Newfoundland with cratonic mainland North America as a test of the proposed 30° rotation of the island (Wegener 1966). New studies, however, are urgently needed both in Newfoundland and elsewhere in the Appalachians and interior North America, partly to fill in major gaps in the early Paleozoic paleomagnetic record for those regions, but also because it has become evident during the past decade that much of the earlier published work on Paleozoic apparent polar wandering relative to North America needs revision. It is now recognized that early and mid-Paleozoic rocks in North America have been widely remagnetized in later geologic time, especially the late Paleozoic. The pervasive effect of remagnetization in Appalachian rocks has been pointed out by Roy *et al.* (1983). Irving and Strong (1984, 1985) showed that Devonian and Lower Carboniferous rocks in Newfoundland carry partial magnetic overprints acquired during the Kiaman reverse magnetic interval (Late Carboniferous to Early Permian) and that Kiaman overprinting affected also many Paleozoic rocks in mainland North America. In carbonates, this problem tends to be compounded by a very low remanence intensity, though a number of successful studies of early and mid-Paleozoic carbonates in North America have been done with cryogenic magnetometers. Although in some of these (e.g., Wisniowiecki *et al.* 1983; McCabe *et al.* 1983, 1984; Elmore *et al.* 1985) there is evidence of complete chemical remagnetization in the late Paleozoic, other carbonate bodies are reported to have escaped remagnetization, either partially (e.g., Dunn and Elmore 1985) or completely (e.g., McCabe *et al.* 1985; Jackson and Van der Voo 1985).

It is of obvious interest to determine whether early Paleozoic

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carbonates on the Western Platform of Newfoundland also may have escaped total remagnetization and in that case to identify the original remanence. The results reported in this paper are from the Lower Ordovician St. George Group and the Middle Ordovician Table Head Group on Port au Port Peninsula and were obtained using a cryogenic magnetometer. Results of two previous studies of the St. George Group were reported by Beales *et al.* (1974) from the Northern Peninsula of Newfoundland and by Deutsch and Rao (1977) from the Port au Port Peninsula, with comparable results. In both investigations, however, spinner megnetometers operating at the limit of resolution were used, so that the remanence may not have been fully resolved in either case. Therefore, we consider the present results from the St. George Group supersede those of Beales *et al.* and Deutsch and Rao. One previous result from the Table Head Group on Port au Port Peninsula was reported as an abstract (Evans and Hall 1985) in which, however, there was no mention of the laboratory treatment used. Still unpublished are recent results (Prasad 1986) from the St. George and Table Head groups on the Northern Peninsula, 170–260 km north of Port au Port.

Geology

The rocks we studied form part of the Humber Zone (Williams 1979). According to Williams and Stevens (1974), the lower Paleozoic rocks of this region record the evolution of a platform at the margin of a proto-Atlantic ocean and its partial subduction under oceanic lithosphere. Rodgers and Neale (1963) recognized two distinct sequences in western Newfoundland: autochthonous shallow-water carbonates and minor siliciclastics, and allochthonous deeper water deposits. The general stratigraphy was summarized by Whittington and Kindle (1969). Within the autochthon, Lower Cambrian siliciclastic rocks are overlain by a Middle Cambrian to Lower Orodvician (Port au Port and St. George groups) shallow-water carbonate platform, which then subsided and received sediments of progressively deeper water origin. This is recorded in the overlying Middle Ordovician rocks (Table Head Group). The following summaries are mainly after James and Stevens (1982) and Klappa *et al.* (1980).

The St. George Group on Port au Port Peninsula (Fig. 1) lies conformably on the Cambrian rocks. Its 573 m sequence of thick-bedded subtidal and peritidal limestone, dolomitic limestone, and dolostone is subdivided into the (1) Watts Bight, (2) Boat Harbour, (3) Catoche, and (4) Aguathuna formations, in ascending order (Table 1). The first two correspond to the former "Isthmus Bay" Formation, which has been provisionally reclassified (Klappa *et al.* 1980), but in our study Watts Bight and Boat Harbour are treated as a single unit (symbol IB, Table 1). The Watts Bight and Boat Harbour formations predominantly consist of interbedded thin to thick limestone and dolostone beds. The Catoche Formation mainly comprises burrowed, fossiliferous limestone. Aguathuna is the thinnest of the four formations and consists of interbedded dolostone and limestone. From cephalopod (Flower 1978) and other fossil evidence, the Watts Bight and Boat Harbour formations are predominantly Gasconadian (early Canadian or Tremadoc) in age, the Catoche Formation is also early Canadian (early Arenig), and the Aguathuna Formation is late Canadian (late Arenig). Therefore the St. George Group falls in the age span 500–475 Ma. The St. George Group is separated from the overlying Table Head Group by an erosional disconformity (Klappa *et al.* 1980), which is well displayed at Aguathuna quarry.

The Table Head Group has been subdivided (Klappa *et al.* 1980) into the (1) Table Point, (2) Table Cove, (3) Black Cove, and (4) Cape Cormorant formations, in ascending order. Our present results are from (1) and (2) only (Table 1, Fig. 1). The combined thickness of the Table Point and Table Cove formations on Port au Port Peninsula is roughly 70 m. The Table Point Formation comprises bioturbated, massive, grey limestones and interbedded limestone–dolostone units. Dolomitic beds are minor components compared with limestone. A typical dark, mottled limestone is well exposed as Aguathuna quarry. The Table Cove Formation is characterized by bioturbated thin- to medium-bedded limestone of possible pelagic origin and by interbedded shales. The rocks are well exposed at West Bay quarry. From graptolite (Finney and Skevington 1979) and other fossil evidence, the age of the Table Head Group is post-Canadian, corresponding to the Whiterock Stage (early Llanvirn, or about 470–465 Ma).

The Cambrian and most of the Ordovician succession on Port au Port Peninsula is gently tilted: dips of the St. George Group and Table Point and Table Cove formations are in the range of 11–18°NW to N. Of the three major orogenic events that affected the Humber Zone, the Taconic (mid- to Late Ordovician) orogeny is very unlikely to be a cause of this deformation (O'Brien 1975). Most probably the deformation resulted from the Acadian (Devonian) orogeny (Schillereff and Williams 1979). The effect of the late Paleozoic Alleghenian orogeny in the Humber Zone is minimal compared with its effect elsewhere in the Appalachians (Williams 1980), though a possible Alleghenian deformation in some lower Paleozoic strata in the Humber Zone cannot be excluded (R. K. Stevens, personal communication, 1986).

Sampling and measurement

Cambrian and Ordovician formations on Port au Port Peninsula were sampled in 1982 and 1984 (Prasad 1986). The initial study reported here is confined to the Ordovician rocks. We used 85 magnetically oriented hand samples of limestone and dolostone from 21 sites at four localities (Fig. 1, Table 1) in the St. George Group (Watts Bight, Boat Harbour, Catoche, and Aguathuna formations, 14 sites) and Table Head Group (Table Point and Table Cove formations, seven sites). Their total stratigraphic span is 500–550 m. The samples were collected from coastal exposures and quarries. The bedding is nearly uniform over the study area. Strike and dip were measured to 1° at 18 sites but could not be measured reliably at the remaining three sites.

A total of 150 standard 2.5 cm cylindrical specimens cut from the 85 samples were used in measurements. Of these, 85 specimens (one from each sample) were used in stepwise thermal demagnetization with a Schonstedt TSD-1 unit operating in residual fields of 5 nT or less. A second set of 47 specimens from 47 samples were given alternating-field (AF) treatment to 100 mT with a Schonstedt GSD-1 unit; steps to 300 mT were sometimes added, using a locally built AF demagnetizer with three-axis tumbler. The remaining 18 specimens were used to study isothermal remanence (IRM) acquired in fields up to 0.86 T and measured with a Schonstedt SSM-1 spinner magnetometer.

IRM apart, all remanence measurements were made with our CTF cryogenic magnetometer. The natural remanent magne-

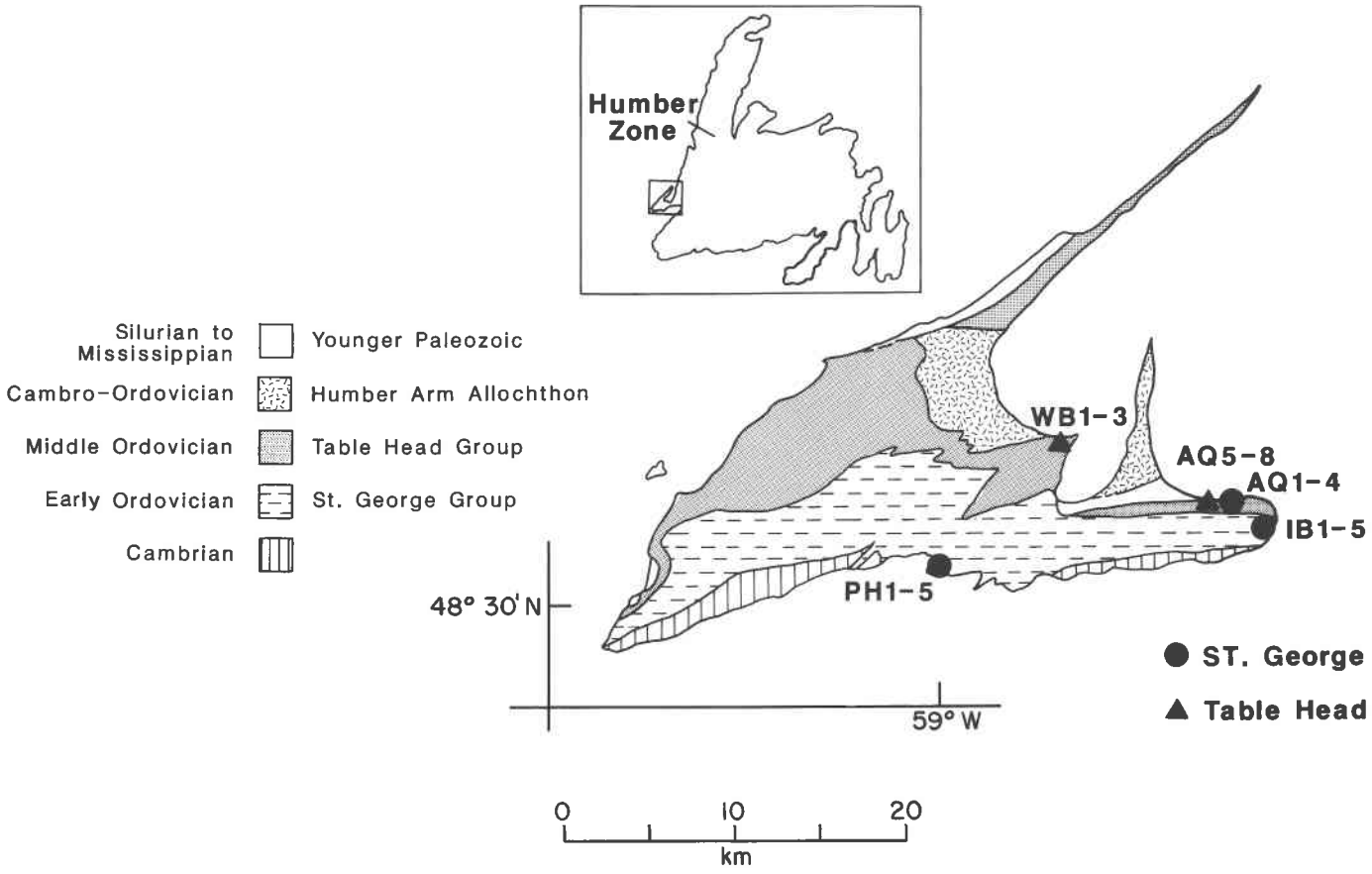


FIG. 1. Simplified geologic map of the study area showing the sampling localities (Table 1) and site numbers.

TABLE 1. Stratigraphic level of the sampling localities

Age	Group	Formation	Locality	Symbol
Middle Ordovician	Table Head	Table Cove	West Bay	WB
		Table Point	Aguathuna quarry	AQ
Early Ordovician	St. George	Aguathuna	Aguathuna quarry	AQ
		Catoche	Pigeon Head	PH
		Boat Harbour and Watts Bight	Isthmus Bay	IB

NOTE: Formation names are as proposed in Klappa *et al.* (1980).

tization (NRM) range of the specimens is 5×10^{-5} to 2×10^{-3} A m^{-1}, or 5×10^{-10} to 2×10^{-8} A m^2 in magnetic moment. As instrumental noise is less than 5×10^{-12} A m^2, all but the weakest specimens remained measurable when demagnetized to 1–2% of NRM. All measurements were corrected vectorially for the specimen holder moment ($\sim 5 \times 10^{-11}$ A m^2), whose effect could be appreciable with very weak specimens.

Results and analysis

Natural remanence

NRM directions for the St. George and Table Head groups are plotted separately in Fig. 2. The two diagrams show similar smeared distributions, with all north poles downward. Their trend led us to suspect that the NRM is the resultant of (1) a steeply inclined component close to the present field (333°, +73°) or axial dipole field (0°, +66°) and (2) at least one other component inclined less steeply to the southeast. This proved to be a good guess.

Pilot study

To identify components, we first selected randomly one sample each from the 21 sites: one specimen from each sample was thermally demagnetized and a second specimen AF demagnetized. Figure 3 shows two examples. Thermal treatment of specimen AQ11-A (Fig. 3*a*) caused the remanence vectors to shift southeastward from NRM into shallower directions roughly along a great circle. Between NRM and 200°C a steep magnetization is being removed. Between 200 and 450°C the successive vectors become well grouped ($D \sim 150°$, $I \sim +25°$), and above 450°C they disperse. Specimen AQ16-C (Fig. 3*c*) responded similarly. In both specimens the remanence has been effectively (96–99%) destroyed by 500°C, so the directions shown for 500°C and higher tempera-

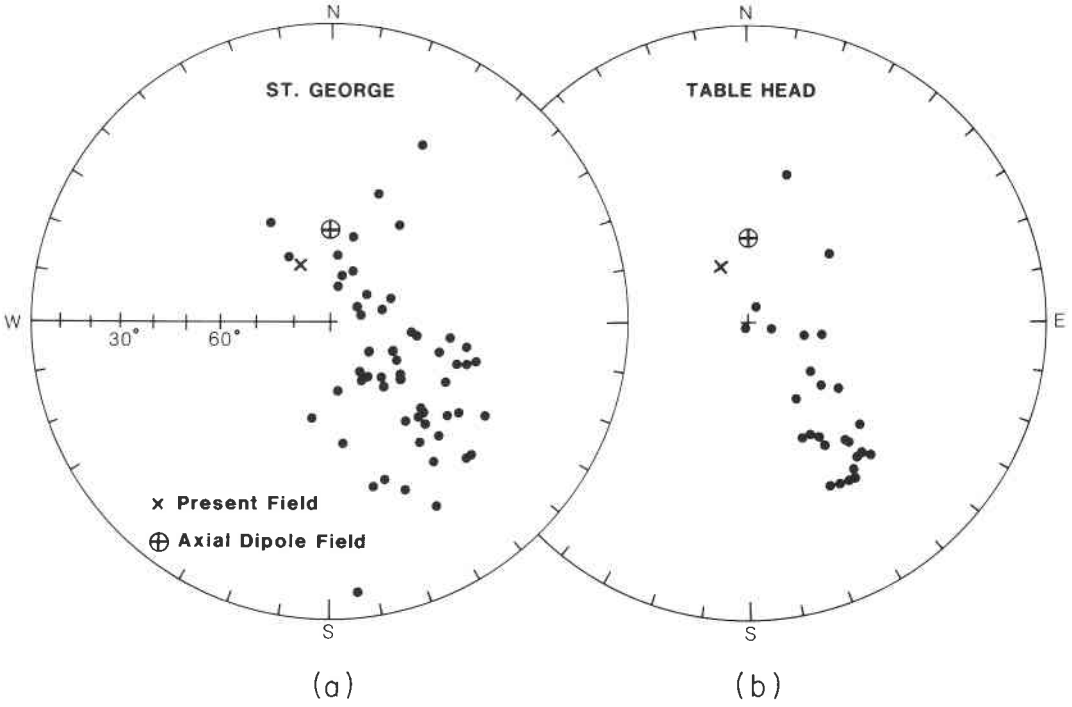


FIG. 2. NRM directions of samples based on one specimen per sample, from (*a*) St. George sites; (*b*) Table Head sites. Lower-hemisphere equal-area projections.

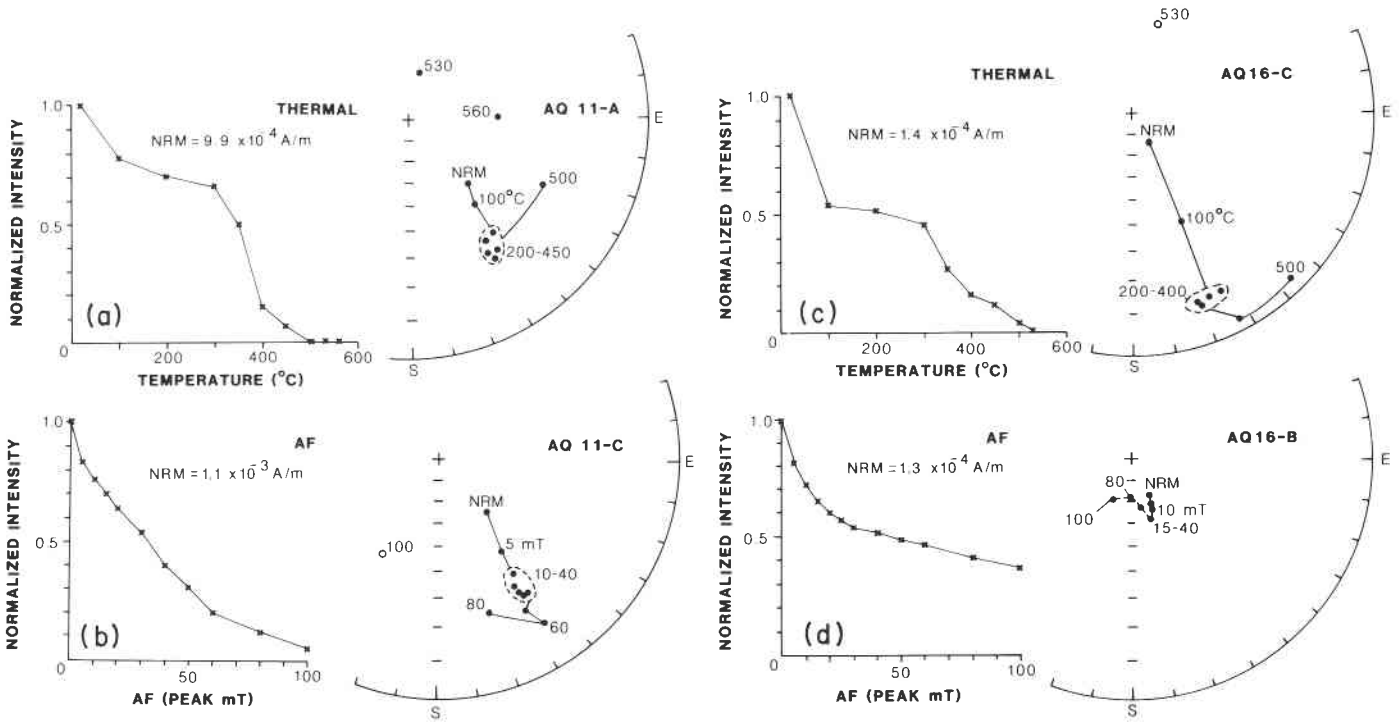


FIG. 3. Demagnetization characteristics of specimen pairs (*a, b* and *c, d*) from two St. George samples after (*a, c*) thermal demagnetization; (*b, d*) AF demagnetization. Solid circles denote lower hemisphere; open circles, upper hemisphere projections on a stereographic (Wulff) net. Dashed outlines indicate directions in the temperature or field intervals shown and have no statistical significance.

tures can be taken as random. Therefore the directions at 400 or 450°C probably are end points of a single component, which we have called the "A" component.

AF treatment affected each of the two samples differently. In one case (Fig. 3*b*) it reproduced the directional changes seen in the thermal treatment (Fig. 3*a*). The mean "end-point" direction at 10–40 mT is identical to the thermally inferred A direction. In the other case (Fig. 3*d*) the AF treatment had little effect on the steep NRM, about 40% of which is left at 100 mT. Sixteen of the 21 thermally treated pilot specimens

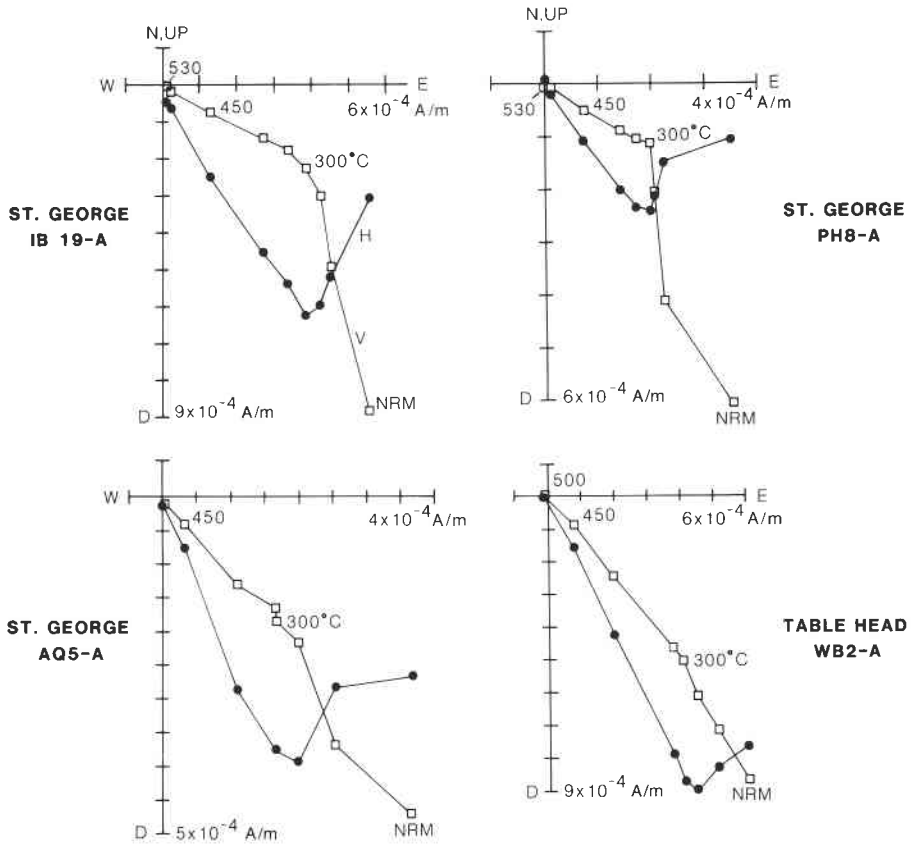


FIG. 4. Representative vector diagrams (Zijderveld 1967) of samples yielding the A component at high temperature. Open symbols are projections of end points of the demagnetization vectors on an east–west vertical (V) plane; solid circles, on the horizontal (H) plane.

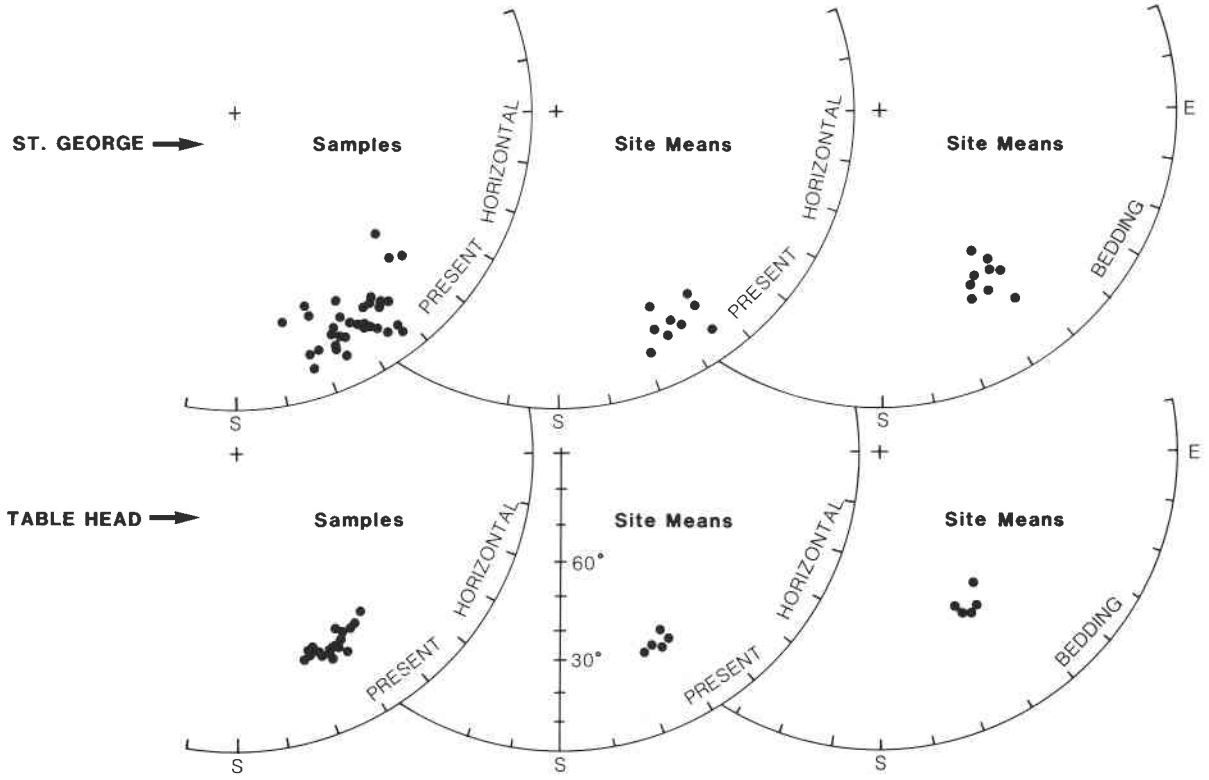


FIG. 5. Characteristic sample and site-mean directions for the A component. Equal-area projection on the lower hemisphere.

TABLE 2. Site-level characteristic directions, A component

Site	n/n_0	*In situ* D (°)	*In situ* I (°)	Tilt-corrected D (°)	Tilt-corrected I (°)	k	α_{95} (°)
St. George Group							
IB5	4/4	154	+17	154	+34	147	8
PH2	3/4	150	+18	150	+30	137	11
PH3	4/4	145	+26	145	+38	32	16
PH4	4/4	145	+21	144	+33	365	5
PH5	4/4	145	+11	145	+23	646	4
AQ1	3/4	155	+28	148	+43	82	14
AQ2	3/4	156	+21	151	+36	242	8
AQ3	4/4	152	+21	146	+35	465	4
AQ4	3/4	159	+14	155	+30	174	9
Table Head Group							
AQ5	4/4	157	+28	154	+42	598	4
AQ8	3/4	151	+32	152	+47	122	11
WB1	4/4	155	+29	153	+39	652	4
WB2	4/4	150	+29	148	+39	308	5
WB3	4/4	153	+27	151	+38	435	4

NOTES: Sites are as shown in Fig. 1; n/n_0, number of sample characteristic directions used in computing the site mean/number of samples thermally demagnetized; D, declination; I, inclination; k, Fisher's precision parameter; α_{95}, radius of the 95% circle of confidence; beds were restored to horizontal by tilting about the strike; dip values are in the range 11–18°N or NW. For details of thermal treatment, see text.

showed direction trends as in Figs. 3*a* and 3*c*. Only nine of the 21 AF-treated specimens matched this behaviour, as in Fig. 3*b*. In the others the remanence either remained steep as in Fig. 3*d* or failed to reveal an end point close to A. Since thermal demagnetization clearly was more effective than AF for resolving these NRM's, we concentrated on thermal treatment from then on.

The A component

Next, 64 specimens, one from each sample not used in the pilot study, were thermally demagnetized. Over 60% clearly revealed an A component in stereograms. All (including pilot) results showing this component were analyzed through orthogonal vector plots (Zijderveld 1967). Figure 4 shows four representative examples, each from a different rock formation. In each, the paired curves show a dual trend: a low-temperature segment depicts the steep magnetization, and a linear or nearly linear high-temperature segment that approaches the origin corresponds to the A component. The A direction was calculated from Zijderveld plots by least-squares fitting. Results were accepted when at least three (usually four or more) approximately colinear points defined a segment terminating near the origin. Out of 51 acceptable samples, 32 were from the St. George Group (nine sites) and 19 from the Table Head Group (five sites). Figure 5 shows the *in situ* sample and site-mean directions along with site means corrected to bedding planes. The various vector groupings are farily compact, and at most sites the within-site precision (k, Table 2) is large. Although all sampling localities are represented in Table 2, the A component is poorly defined or absent at five sites (IB1–IB4, PH1) of the lower St. George sequence and at two Table Head sites (AQ6, AQ7). The overall tilt-corrected A directions (Table 3) for St. George sites (148.5°, +33.6°, α_{95} = 4.3°) and Table Head sites (150.0°, + 40.7°, α_{95} = 3.9°) are similar. For both the St. George and Table Head groups the directions calculated from site unit vectors are identical to those based on samples. The site-based statistics are used because their (larger) associated errors are probably more realistic, and their use is justified because at least three samples were averaged per site.

Out of the 47 AF-demagnetized specimens (including pilots), 22 specimens supported the thermal results (e.g., Fig. 3*b*), but since the other 25 specimens did not yield an A component, the AF results were not analyzed further.

The B component

In 14 samples, a south- to southeast-directed remanence with low, usually negative inclination was isolated by the thermal treatment. An example (Fig. 6*a*) shows that first a steep magnetization is removed, similar to what occurred in specimens carrying the A component. Between 300 and 530°C, the remanence stabilizes around a mean end-point direction ($D \sim 165°$, $I \sim -10°$). By 560°C, 98% of the NRM is destroyed and directions become random. Zijderveld plots (Fig. 6*a*, right) display this trend in two linear segments, the second of which goes to the origin. We conclude that the 300–530°C cluster in the stereonet represents a single component, which we call "B". Figure 6*b* shows one of only two cases where AF treatment confirmed a B direction.

All 14 B samples are from the St. George Group. All but one are from the same five lower sites (IB1–IB4, PH1) at which the A component is poorly defined or absent. The remaining sample, AQ6 (from a higher exposure), provided the only example where A and B can be separately identified in the same rock (Figs. 7*b*, 7*c*). The A component is erased in the range 300–450°C (Fig. 7*c*); then a stable B direction is obtained at high temperatures (Fig. 7*b*). All other successfully demagnetized samples showed the presence of either A or B alone, along with the steep component. These findings suggest that either the A or the B component had once been omnipresent in the rocks and, at certain sites, was selectively replaced by the other component when the ambient field had changed. Later we show it is B that has overprinted A.

Seven of the 14 B directions (Fig. 8) were calculated from Zijderveld plots as before. In the other seven specimens the plots are erratic near the origin, so the mean directions of well-grouped ($k \geq 70$) end points were used instead. Since the B component occurs too sparsely for averaging by sites, its mean direction was calculated from the 14 sample unit vectors. The *in situ* direction is 167.9°, −9.2°, α_{95} = 7.0° (Table 3).

The steep component

We saw that all NRM's tend to be dominated by a steep remanence. Since this steep, viscous component always occurs with at least one of components A and B, it is plausible that it postdates both A and B. Being steep, the viscous component may be of relatively recent origin.

Magnetic mineralogy

In order to identify magnetic minerals we measured the IRM of 18 specimens from 17 sites. They form two sets, depending on whether the thermally treated specimens from the same samples terminated in the A or the B component. Three kinds of curves were observed (Fig. 9); they are similar to IRM curve types reported from limestones elsewhere (Lowrie and Heller 1982). Figure 9*a* is characteristic of magnetite as the sole remanence carrier, whereas Fig. 9*c* is typical of hematite.

TABLE 3. Final statistics after thermal demagnetization

Mean	Fisher statistics					Antipole position				
	D (°)	I (°)	k	N	α_{95} (°)	λ_p (°)	Lat. (°N)	Long. (°E)	dp (°)	dm (°)
St. George Group										
A component, 9 sites, *in situ*	150.9	19.6	120	9	4.7	10.1 S				
A component, 9 sites, tilt-corrected	*148.5*	*33.6*	*144*	*9*	*4.3*	*18.4 S*	*17.5*	*152.3*	*2.8*	*4.9*
A component, 32 samples, tilt-corrected	148.2	33.4	80	32	2.9	18.3 S				
B component, 14 samples, *in situ*	*167.9*	*−9.2*	*33*	*14*	*7.0*	*4.6 N*	*44.9*	*138.2*	*3.6*	*7.1*
Table Head Group										
A component, 5 sites, *in situ*	153.3	29.0	623	5	3.1	15.5 S				
A component, 5 sites, tilt-corrected	*150.0*	*40.7*	*394*	*5*	*3.9*	*23.3 S*	*13.4*	*149.2*	*2.9*	*4.7*
A component, 19 samples, *in situ*	153.4	28.8	273	19	2.0	15.4 S				
A component, 19 samples, tilt-corrected	150.3	40.5	232	19	2.2	23.1 S				

NOTES: N, number of sites averaged; λ_p, paleolatitude; dp, dm, semi-axes of 95% confidence oval. Other symbols as in Table 2.

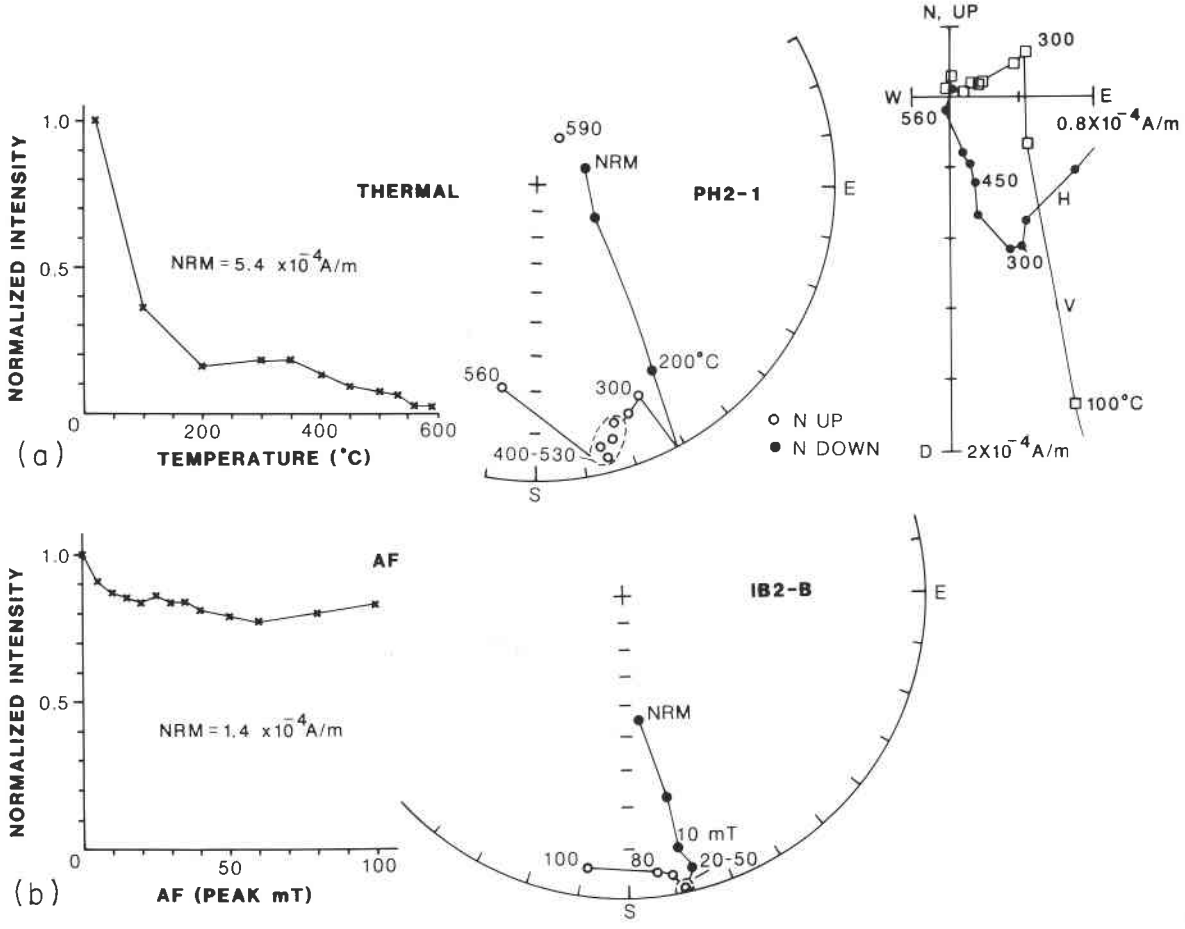


FIG. 6. Demagnetization characteristics of two St. George samples yielding B component. (*a*) Thermal results plotted on a stereonet and as a vector diagram; (*b*) AF results plotted on stereonet only. All symbols as in Figs. 3 and 4.

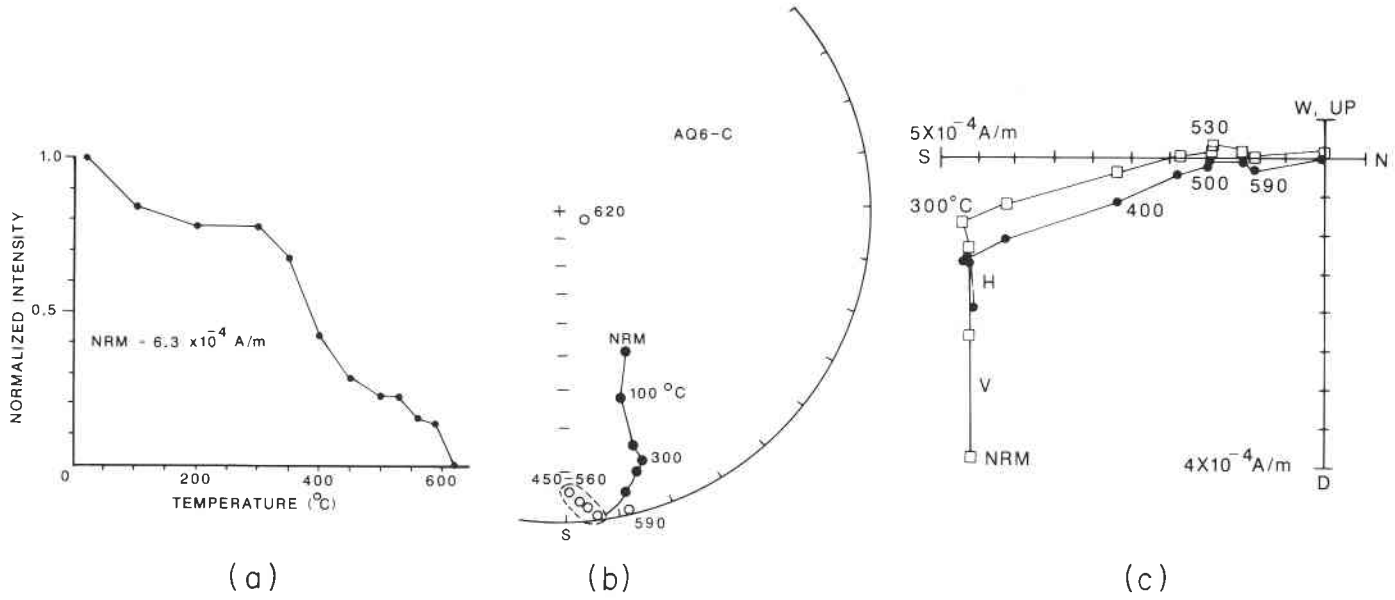


FIG. 7. Thermal demagnetization results of the St. George sample yielding both A and B components as inferred from (*b*) stereoplot and (*c*) vector diagram. All symbols as in Figs. 3 and 4.

Eight of 14 specimens from A-component samples gave IRM curves as in Fig. 9*a*, indicating that in these samples both A and the steep viscous component are magnetite. This is consistent with the thermal demagnetization results, which showed invariably (e.g., Fig. 3) that the A component has become unblocked at temperatures below 500 or 530°C. The six other A specimens gave "intermediate" IRM curves (Fig. 9*b*), suggesting that magnetite and hematite occur together. This would mean that the hematite fraction probably did not contribute significantly to the NRM. Four samples carrying the B component gave "hematite" curves (Fig. 9*c*). Hematite is consistent with demagnetization results of B-component specimens, some of which remained blocked beyond 600°C (Fig. 7), while their slow AF decay (Fig. 6) is evidence of considerable hardness.

Hematite type IRM curves could be diagnostic also of goethite (Lowrie and Heller 1982). Earlier in this study we did, in fact, suspect a dominant goethite presence in the NRM of two samples whose aberrant response to thermal treatment included a 96% loss of remanence below 100°C. Their NRM directions also were atypical. These results, being clearly anomalous, were discarded, and we consider it safe to rule out goethite as a possible serious contaminant in all other samples. We conclude that the A component is always magnetite and the B component always hematite.

Discussion

The B component is of reverse polarity. Its *in situ* direction (167.9°, −9.2°), when inverted, yields a north pole at 44.9°N, 138.2°E; dp, dm = 3.6°, 7.1° (Fig. 10, pole B), corresponding to a low paleolatitude ($\lambda_p = 5 \pm 4$°N) for Port au Port Peninsula. Pole B is statistically identical to the H pole (Fig. 10, pole DLH; $\lambda_p = 7 \pm 5$°N) isolated by Irving and Strong (1984) in the Early Carboniferous Deer Lake Group in western Newfoundland and considered by them an early Kiaman overprint on an original Viséan remanence. Another case of Kiaman overprinting on Early Carboniferous rocks in western Newfoundland was reported by Murthy (1985). Irving and Strong (1984, 1985) noted that characteristically reversed

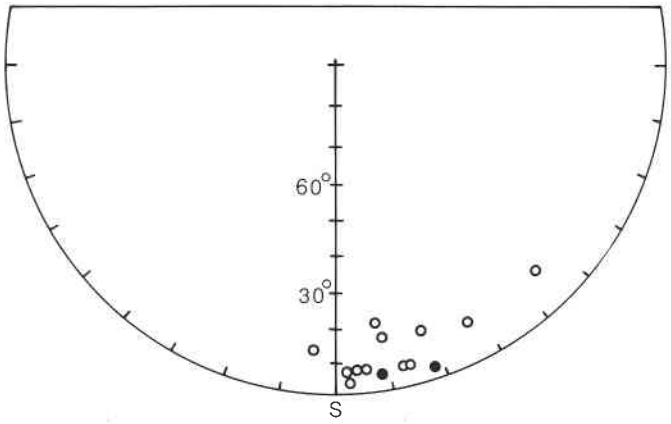


FIG. 8. Sample characteristic directions of the B component after thermal demagnetization. Equal-area projection. Symbols as in Fig. 3.

Kiaman magnetizations with near-equatorial paleolatitudes as in Deer Lake H occur widely as overprints in Late Devonian and Early Carboniferous rocks of cratonic mainland North America (Fig. 10) and also in the Newfoundland Avalon Zone.

We conclude that the St. George B component is also of Kiaman age. As the main deformation on Port au Port Peninsula probably was pre-Kiaman, we have not corrected the B direction for tilt. The B remanence (hematite) is probably of chemical origin, since a predominant thermal remagnetization seems to be precluded by the very low value of the conodont alteration index (CAI = 1) of the Port au Port rocks (Nowlan and Barnes 1987); this value corresponds to a maximum burial temperature of about 80°C (Epstein *et al.* 1977).

The St. George carbonates are the oldest rocks in Newfoundland so far reported to carry a Kiaman remanence. While this attests to the pervasiveness of late Paleozoic remagnetization, it is important to note that *most* of the rocks in the present study do not have a B component and so they appear to have escaped remagnetization at least during the Kiaman interval.

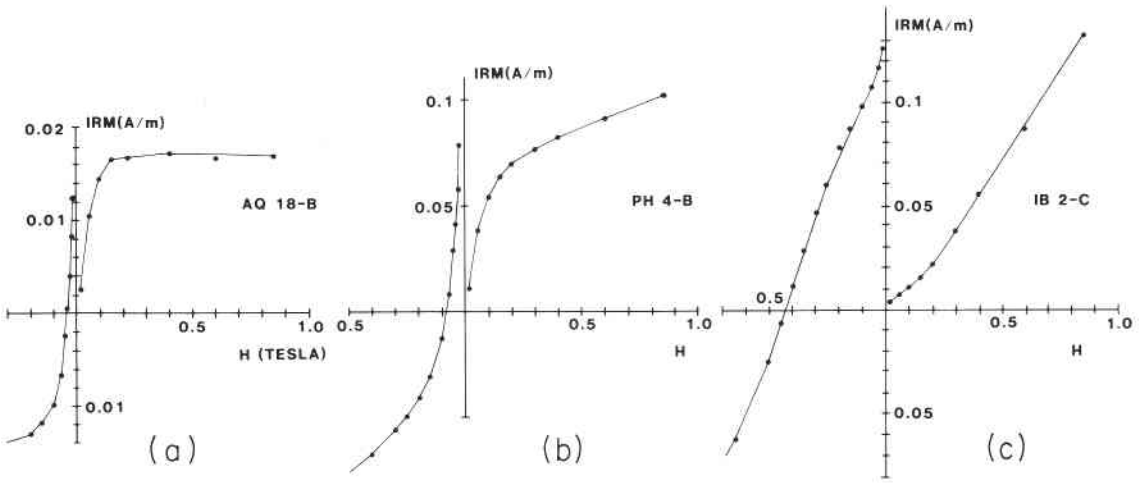


FIG. 9. Isothermal remanent magnetization (IRM) acquisition and back-field characteristics. Curves in (*a*) and (*b*) are from samples yielding the A component, whereas the curve in (*c*) is representative of samples giving the B component after thermal demagnetization.

The A component (Table 3) points southeast and downward, indicating it has reverse polarity like the St. George B component. The *in situ* A directions for both St. George (150.9°, +19.6°) and Table Head (153.3°, +29.0°), however, deviate significantly from the B direction (167.9°, −9.2°), showing that A is not of Kiaman age. A later age is very unlikely, since post-Paleozoic field directions relative to western Newfoundland should be typically much steeper than the observed A directions. The A component also is misaligned with reported Early Carboniferous directions from western Newfoundland (Irving and Strong 1984; Murthy 1985). All this indicates that the A component is pre-Carboniferous.

We interpret the A component as an original Orodovician magnetization for the following reasons. First, as demonstrated, the A component resides in magnetite, and therefore (Lowrie and Heller 1982) the characteristic remanence was probably acquired at or shortly after deposition of the limestone. Second, it was noted that whenever our thermal treatment had terminated in the A component, all remanence was lost on further heating, showing the absence of additional components. Third, our previous argument, that because of the very low CAI value (~1) in the Port au Port area the B component cannot be mainly a thermal remagnetization, applies also to the A component.

These findings are consistent with a primary or early diagenetic age of the A component but are insufficient to prove it. An alternative is chemical remagnetization. Scotese *et al.* (1982), Wisniowiecki *et al.* (1983), and McCabe *et al.* (1983, 1984) reported stable secondary magnetizations in magnetite in early and mid-Paleozoic carbonates in eastern North America, which they attributed to late Paleozoic overprinting through newly developed magnetite. In all these studies the overprint directions are characteristically Kiaman, in sharp contrast to our (pre-Kiaman) A-component directions. However, there seem to be no reported examples of North American carbonates carrying a pre-Kiaman chemical overprint in magnetite. These facts conspire to weaken, though they do not rule out, the case for a secondary chemical origin of the A component.

Let us now compare the A component with published Ordovician results from cratonic North America. To this end we use the tilt-corrected directions for the St. George and Table Head groups (148.5°, +33.6° and 150.0°, + 40.7°; Table 3); the assumption is that for both groups the remanence predates the (Acadian?) deformation. A proper fold test for testing this assumption is not possible because bedding attitudes are uniform in the study area except for small variations in strike and dip between sites. As a result of the latter variations, the nine St. George site directions are slightly better grouped after than before tilt correction (compare values of precision k, Table 3), whereas the five Table Head sites are more poorly grouped, but in neither case is the change in k large enough for an adequate test. For St. George, the tilt-corrected A direction, when inverted, yields a north pole at *17.5°N, 152.3°E; dp, dm = 2.8° 4.9°* (Fig. 10, pole A1). For Table Head, the north pole is at *13.4°N, 149.2°E; dp, dm = 2.9°, 4.7°* (pole A2).

Poles A1 and A2 are compared with five of the more reliable published cratonic poles cited as Ordovician (Fig. 10, circles). The seven poles form (1) an "eastern" set, which includes A1 and A2, located about 10−35°N, 150°E (except pole OD: 166°E); and (2) a "northwestern" pair close to 30°N, 115°E. One of the eastern poles (OD) is from carbonates in the interior part of the craton. Poles A1, A2, and OD (Oneota dolomite, Lower Ordovician) are situated at low (10−20°) latitudes, far from late Paleozoic poles, but are close to a cluster of Cambrian, including some Late Cambrian, North American poles centered on the equator (e.g., Fig. 12 in Dankers and Lapointe 1981). Poles A1 and A2 also lie near the "syenite" pole (18°N, 142°E), dated at 535 Ma from the Cambro-Ordovician McClure Mountain complex of Colorado (Lynnes and Van der Voo 1984). The four "northerly" (~30°N) Ordovician poles (LP, MB, CR, J) are from Appalachian studies. They include one Newfoundland result, pole LP from the uppermost Middle to Upper Ordovician Long Point Group on Port au Port Peninsula (Murthy 1983). Pole LP lies 10−15° north of our poles A1 and A2 from the older Ordovician rocks on Port au Port Peninsula. Poles MB and CR (Moccasin-Bays and Chapman Ridge formations) are both based on mid-Orodovician rocks from the Valley and Ridge Province, but they are about 30° apart. Watts and Van der Voo (1979) offered two possible explanations for this discrepancy, one being that poles MB and CR both represent original magnetizations, that of CR being older. This requires an apparent polar wandering (APW) loop from CR to MB and back to the vicinity to the Late Ordovician pole J (Juniata Formation). Such a loop, however, is not

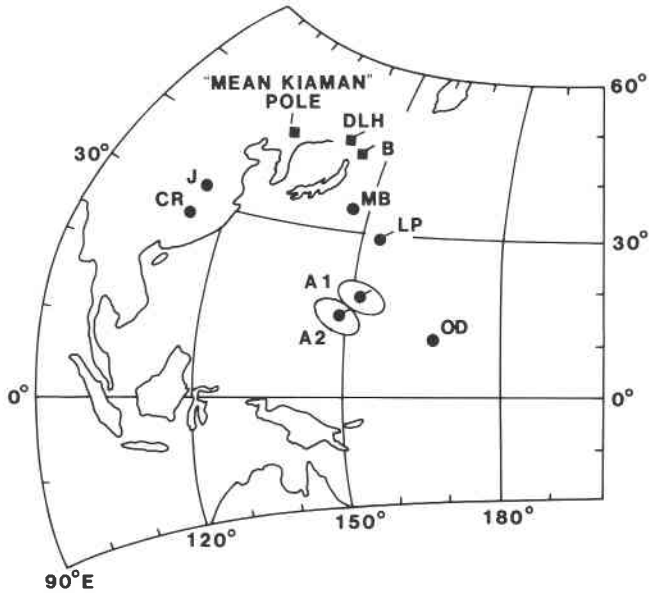


FIG. 10. Paleopoles from this study plotted with selected Ordovician and late Paleozoic poles for North America. Circles, poles cited as Ordovician; squares, poles cited as Late Carboniferous to Permian ("Kiaman"). Symbols with bar, Newfoundland poles; without bar, mainland poles. Ordovician poles: A1, with 95% confidence oval, St. George A component (this study); A2, with oval, Table Head A component (this study); CR, Chapman Ridge Formation, eastern Tennessee (Watts and Van der Voo 1979); J, Juniata Formation, Virginia and Pennsylvania (Van der Voo and French 1977); LP, Long Point Formation, Port au Port Peninsula, Newfoundland (Murthy 1983); MB, Moccasin-Bays, eastern Tennessee (Watts and Van der Voo 1979); OD, Oneota dolomite, upper Mississippi River valley (Jackson and Van der Voo 1985). Kiaman poles: B, St. George B component (this study); DLH, Deer Lake Group H component (Irving and Strong 1984). "Mean Kiaman" pole, based on three mainland Early Carboniferous studies cited by Irving and Strong (1984): this is identical to an "early Kiaman" pole based on data for the interval 265–295 Ma (Irving and Irving 1982).

justified by the more recent data (Fig. 10), which include Middle and Late Ordovician pole positions among the eastern set south of pole MB. The explanation favoured by Watts and Van der Voo was that pole CR, being close to pole J, may actually represent a Late Ordovician or Silurian field. Relevant here are the results of two studies of the Middle Ordovician Trenton Limestone, the first with a pole at 36°N, 114°E (McElhinny and Opdyke 1973). McCabe *et al.* (1984) obtained a different Trenton pole, near typical "Kiaman" poles, and showed that the remanence direction corresponding to McElhinny and Opdyke's pole is actually the resultant direction of a late Paleozoic overprint and the present Earth's field. Because poles CR and J lie near the old Trenton pole and because this northwestern cluster is relatively close to the "Kiaman" poles, it is possible that poles CR and J both represent post-Ordovician remagnetizations. Poles CR and J are fairly close also to two pole positions not shown in Fig. 10, reported from St. George Group carbonates in the Northern Peninsula of Newfoundland (26°N, 126°E, Beales *et al.* 1974; 21°N, 113°E. Prasad 1986). Both are based on normal magnetizations, whereas only reverse polarity was found by us on Port au Port Peninsula. As was noted earlier, the Beales *et al.* pole may be based on an unresolved remanence. In the area of the St. George Group exposures studied by Beales *et al.* and Prasad, conodont CAI values are in the range of 2–3.5 (Nowlan and Barnes 1987), making it probable that these carbonates have been thermally remagnetized since the Ordovician (Prasad 1986).

We conclude that the low-latitude poles (A1, A2, OD) are more representative of the Early and mid-Ordovician geomagnetic field relative to the craton than are the northwestern poles, and further, that poles A1 and A2 truly represent the prevailing field directions on Port au Port Peninsula when the St. George and Table Head carbonates were deposited. These two results are thus additions to what is still a very small set of dependable Ordovician data for cratonic North America. The fact that poles A1 and A2 are close suggests that little if any polar movement occurred relative to western Newfoundland between Early and mid-Ordovician time. We are unable to explain the wide separation between the two Middle Ordovician poles A2 (Newfoundland) and MB (mainland).

Western Newfoundland probably was in low southern latitudes during much of the Paleozoic. The average paleolatitude (λ_p) is about 20°S for the Early and Mid-Ordovician (this study, Table 3), 8°S for the Late Ordovician (Murthy 1983), 12–20°S for the Early Carboniferous (Irving and Strong 1984; Murthy 1985), and 5–7°N for the Late Carboniferous to Early Permian (Irving and Strong 1984; this study). The changes in λ_p are somewhat erratic, and the Late Ordovician minimum is of uncertain significance.

Rotation of western Newfoundland

This discussion bears on Wegener's (1966) hypothesis that Newfoundland has rotated 30° anticlockwise relative to the mainland. Pole A1 (Fig. 10) supersedes a previous result (Deutsch and Rao 1977) for the St. George Group on Port au Port Peninsula. Deutsch and Rao compared this earlier result with McElhinny and Opdyke's (1973) Trenton Limestone result as part of a test of the postulated rotation, which also included pre-Ordovician comparisons. The remanence vectors for St. George and Trenton both have south declinations, and the two poles (30°N, 119°E; 36°N, 114°E) agree. Deutsch and Rao (1977) interpreted this as a negative test of the proposed rotation of Newfoundland. For the test to be positive, the declination of the St. George samples should have been southeast or northwest, assuming the observed south declination for Trenton. Ironically, both the 1973 Trenton and 1977 St. George results are now obsolete. The new Newfoundland results (St. George and Table Head A component, Table 3; and Long Point Group, Murthy 1983) all do have southeast or northwest declinations. However, the mainland Ordovician poles in the eastern set (Fig. 10) also represent southeast–northwest or east–west declinations. It is easy to show that "correcting" for the assumed rotation of the island would displace the three Newfoundland poles (A1, A2, LP) in Fig. 10 nearly 30° westward, thus greatly increasing their discordance with the mainland poles (OD, MB). Therefore, the recent Ordovician paleomagnetic evidence again fails to support a (post-Ordovician) rotation of western Newfoundland.

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